Density/Temperature Based Wine ABV Calculator

Abstract

This prototype presents a conceptual system for estimating wine alcohol content using density and temperature measurements. A load cell measures the weight of a full bottle, while a temperature sensor accounts for changes in density. A rotary selector allows the user to choose a bottle type (normal, sparkling etc) with a preset empty weight. Using these inputs, the system estimates density and converts it to ABV, displaying the result on a digital screen.

Challenges

Wine is not composed of only water and alcohol; sugars, acids, and other dissolved compounds also affect density, which reduces accuracy. Bottle weights vary between manufacturers, so using preset values introduces estimation error. The system also assumes a full 750 mL bottle, meaning partially filled bottles will produce incorrect results.

Temperature affects density as well, and while a correction is applied, it is only an approximation. Additionally, because this design is conceptual and not physically calibrated, real-world performance may differ. These factors make the system better suited for estimation rather than precise measurement.

```
```C++
selected_bottle_type = read_rotary_encoder()
preset_bottle_mass_g = get_bottle_preset(selected_bottle_type)

total_bottle_mass_g = read_load_cell()
temp_c = read_temperature()

liquid_mass_g = total_bottle_mass_g - preset_bottle_mass_g
if liquid_mass_g < 0
    liquid_mass_g = 0

density = liquid_mass_g / 750.0

abv = convert_density_to_abv(density, temp_c)

display_bottle_type(selected_bottle_type)
display_abv(abv)
```
```

```
float convert_density_to_abv(float densityMeasured, float tempC) {
  // Step 1: compensate density to 20 °C
  const float beta = 0.00030;  // empirical temperature coefficient, tune expe
  float rho20 = densityMeasured + beta * (tempC - 20.0);

  // Step 2: compute density difference from water at 20 °C
  float d = 0.99820 - rho20;

  // Step 3: quadratic fit for approximate ABV
  float abv = 19399.7 * d * d + 642.6 * d - 0.050;

  // Clamp output to a sensible range
  if (abv < 0.0) abv = 0.0;
  if (abv > 20.0) abv = 20.0;   // this fit is intended for roughly 0-20% ABV

  return abv;
}
```

```
float compensateDensityTo20C(float densityMeasured, float tempC) {
  const float beta = 0.00030;  // g/mL per °C, tunable empirical coefficient
  return densityMeasured + beta * (tempC - 20.0);
}
```

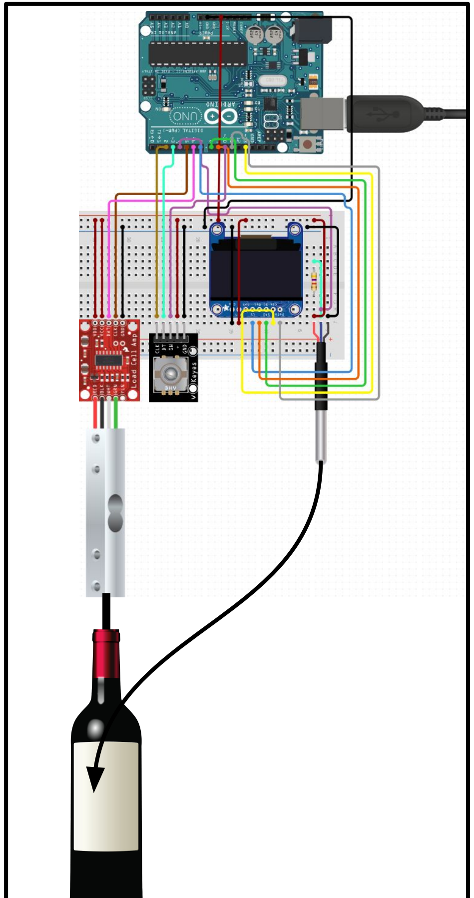
Psuedo code - not representative of final design

How does it function?

The prototype employs two primary components: a load cell for measuring bottle weight and a temperature sensor for determining the wine's ambient temperature. ABV calculation currently relies on the assumption of an average, full bottle size, with the specific weight determined by the bottle type selected via a rotary switch. The full bottle weight is transmitted to the Arduino, which then subtracts the bottle weight to isolate the wine's weight. Finally, the wine's weight is temperature-compensated for precision, followed by an empirical ABV analysis to identify the alcohol content.

Takeaway

The system estimates wine alcohol content (ABV) using a simple, low-cost approach based on mass and temperature measurements. Although currently a conceptual design, it demonstrates strong potential. Future improvements could include enhanced calibration, more precise inputs, and incorporating additional sensing methods (e.g., sugar or volume). This accessible system offers a method for understanding wine properties, though it is not designed for highly precise analytical results.



Prototype schematic - design not final